



AWS Audit Report

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Report Date	2026-08-05
Tests	23
Passed	12 (52.2%)
Failed	11 (47.8%)
Scope	AWS Account ID: 798094440826 In-Scope Regions: ['us-east-1']

Test Summary

Test	Result	Risk	Comments
aws-iam-004: IAM users with an active console password have MFA enabled.	Fail	High	- Population: 12 - Passing: 1 (8.3%) - Failing: 11 (91.7%)
aws-iam-005: Active IAM user access keys are rotated at least every 90 days.	Fail	High	- Population: 7 - Passing: 6 (85.7%) - Failing: 1 (14.3%)
aws-s3-002: S3 buckets are configured to block public access.	Fail	Medium	- Population: 5 - Passing: 2 (40.0%) - Failing: 3 (60.0%)
aws-efs-001: EFS volumes must have default encryption enabled in each region.	Fail	Medium	- Population: 1 - Passing: 0 (0.0%) - Failing: 1 (100.0%)
aws-efs-002: EFS volumes are encrypted at rest.	Fail	Medium	- Population: 1 - Passing: 0 (0.0%) - Failing: 1 (100.0%)
aws-s3-004: S3 buckets must have required tags applied and tag values must not be empty.	Fail	Low	- Population: 5 - Passing: 3 (60.0%) - Failing: 2 (40.0%)
aws-ec2-001: EC2 instances must have required tags applied and tag values must not be empty.	Fail	Low	- Population: 1 - Passing: 0 (0.0%) - Failing: 1 (100.0%)

aws-ec2-001: EC2 security groups have required tags applied and tag values are not empty.	Fail	Low	- Population: 3 - Passing: 0 (0.0%) - Failing: 3 (100.0%)
aws-ebs-003: EBS volumes must have required tags applied and tag values must not be empty.	Fail	Low	- Population: 1 - Passing: 0 (0.0%) - Failing: 1 (100.0%)
aws-lambda-001: Lambda functions must have required tags applied and tag values must not be empty.	Fail	Low	- Population: 3 - Passing: 0 (0.0%) - Failing: 3 (100.0%)
aws-s3-003: S3 buckets are configured to deny unencrypted data in-transit.	Fail	Informational	- Population: 5 - Passing: 1 (20.0%) - Failing: 4 (80.0%)
aws-iam-001: Root account does not have any active access keys.	Pass	High	
aws-iam-003: Root account has MFA enabled.	Pass	High	
aws-rds-002: RDS instances are configured to block public access.	Pass	High	
aws-cloudtrail-001: At least one multi-region CloudTrail trail has logging enabled.	Pass	High	
aws-guardduty-001: GuardDuty is enabled for all in-scope regions.	Pass	High	- Population: 1 - Passing: 1 (100.0%)
aws-iam-006: IAM passwords comply with the organization's password policy.	Pass	Medium	
aws-s3-001: S3 buckets are encrypted at rest.	Pass	Medium	- Population: 5 - Passing: 5 (100.0%)
aws-rds-001: RDS instances are encrypted at rest.	Pass	Medium	
aws-rds-005: RDS instances have deletion protection enabled at the cluster or instance level.	Pass	Medium	
aws-rds-004: RDS instances have automatic minor version upgrades enabled.	Pass	Low	
aws-rds-003: RDS backups are retained for at least 14 days.	Pass	Low	
aws-rds-006: RDS instances must have required tags applied and tag values must not be empty.	Pass	Low	

BACK TO TEST SUMMARY

Test ID	aws-iam-004
Test Description	IAM users with an active console password have MFA enabled.
Risk Rating	High
Test Procedures	1. Obtained a list of IAM users by calling the list_users() boto3 command. 2. Saved the list of users: iam/users.json. 3. For each user, obtained the login profile by calling the get_login_profile() boto3 command. 4. For each user with a login profile, obtained MFA devices by calling the list_mfa_devices() boto3 command. 5. Inspected the MFA device configuration to determine if at least one MFA device is assigned.
Test Attributes	• Users with a login profile have at least one MFA device.
Conclusion	Fail
Comments	Exceptions Noted. 11 of 12 IAM user(s) have an active console password but do not have MFA enabled.

User Name	Result	Comments
goose	Fail	Console password enabled but no MFA device assigned.
iceman	Fail	Console password enabled but no MFA device assigned.
maverick	Fail	Console password enabled but no MFA device assigned.
slider	Fail	Console password enabled but no MFA device assigned.
student_1	Fail	Console password enabled but no MFA device assigned.
student_2	Fail	Console password enabled but no MFA device assigned.
student_2qu2i	Fail	Console password enabled but no MFA device assigned.
student_48tpz	Fail	Console password enabled but no MFA device assigned.
student_hx8sc	Fail	Console password enabled but no MFA device assigned.
student_p6qiq	Fail	Console password enabled but no MFA device assigned.
student_pdfbx	Fail	Console password enabled but no MFA device assigned.

itauditguy	Pass	
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BACK TO TEST SUMMARY

Test ID	aws-iam-005
Test Description	Active IAM user access keys are rotated at least every 90 days.
Risk Rating	High
Test Procedures	1. Obtained a list of IAM users by calling the list_users() boto3 command. 2. Saved the list of IAM users: iam/users.json. 3. For each IAM user, obtained access key metadata by calling the list_access_keys() boto3 command. 4. For each IAM user, saved access key metadata: iam/users/[user_name]/access_keys.json 5. Inspected the 'AccessKeyMetadata' for each user to determine if they comply with the test attribute(s) below.
Test Attributes	• 'CREATE_DATE' <= 90 days ago (for keys with an 'ACTIVE' status).
Conclusion	Fail
Comments	Exceptions Noted. 1 of 7 IAM key(s) are active and over 90 days old.

User	Access Key ID	Result	Comments
itauditguy	AKIA3TURCXF5GCAELVHX	Fail	Key is 1290 days old.
audit-user	AKIA3TURCXF5DUSNOI7O	Pass	
audit-user	AKIA3TURCXF5GOCH6KMT	Pass	
github-actions-itauditguy-deploy	AKIA3TURCXF5DT3R57CB	Pass	
student_1	AKIA3TURCXF5B3HBOKL3	Pass	
student_2	AKIA3TURCXF5A65PHQYT	Pass	
student_48tpz	AKIA3TURCXF5N73Q5YB5	Pass	

BACK TO TEST SUMMARY

Test ID	aws-s3-002
Test Description	S3 buckets are configured to block public access.
Risk Rating	Medium
Test Procedures	1. Obtained a list of S3 buckets by calling the list_buckets() boto3 command. 2. Saved the list of buckets: s3/buckets.json. 3. For each bucket, obtained the public access block settings by calling the get_public_access_block() boto3 command. 4. For each bucket, saved the public access block settings: s3/buckets/[bucket_name]/public_access_block.json. 5. For each bucket, inspected the public access block settings to determine if they comply with the test attribute(s) below.
Test Attributes	• BlockPublicAcls, IgnorePublicAcls, BlockPublicPolicy, and RestrictPublicBuckets are set to true.
Conclusion	Fail
Comments	Exceptions Noted. 3 of 5 S3 bucket(s) are not configured to block public access.

Bucket Name	Result	Comments
bad-bucket-54321	Fail	No Public Access Block configuration found.
itauditguy	Fail	One or more public access settings are disabled.
taylor-swift-secret-vault	Fail	One or more public access settings are disabled.
aws-cloudtrail-logs-798094440826-aa81e64c	Pass	
config-bucket-798094440826	Pass	

BACK TO TEST SUMMARY

Test ID	aws-ebs-001
Test Description	EBS volumes must have default encryption enabled in each region.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained the EBS default encryption settings by calling <code>get_ebs_encryption_by_default()</code> boto3 command. 2. For each in-scope region, saved the EBS default encryption settings: <code>ec2/[region_name]/default_ebs_encryption.json</code>. 3. Inspected the configuration for each region to determine if 'EbsEncryptionByDefault' is set to True.
Conclusion	Fail
Comments	Exceptions Noted. 1 of 1 region(s) do not have EBS default encryption enabled.

Region	Result	Comments
us-east-1	Fail	EBS default encryption is not enabled in this region.

BACK TO TEST SUMMARY

Test ID	aws-ebs-002
Test Description	EBS volumes are encrypted at rest.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained a list of EBS volumes by calling describe_volumes() boto3 command. 2. For each in-scope region, saved the list of EBS volumes: ec2/[region_name]/volumes.json. 3. For each EBS volume, inspected the 'Encrypted' attribute to determine it is set to 'true'.
Conclusion	Fail
Comments	Exceptions Noted. 1 of 1 EBS volume(s) are not encrypted.

Region	Volume ID	Result	Comments
us-east-1	vol-0fad5b6c1c9198060	Fail	EBS volume is not encrypted.

BACK TO TEST SUMMARY

Test ID	aws-s3-004
Test Description	S3 buckets must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. Obtained a list of S3 buckets by calling the list_buckets() boto3 command. 2. Saved the list of buckets: s3/buckets.json. 3. For each bucket, obtained its tags by calling the get_bucket_tagging() boto3 command. 4. For each bucket, saved the tags: s3/buckets/[bucket_name]/tags.json. 5. For each bucket, inspected the tags to determine if the following tag keys exist and have non-empty values: ['Owner', 'Description', 'Classification']
Conclusion	Fail
Comments	Exceptions Noted. 2 of 5 S3 bucket(s) are missing required tags or have empty values.

Bucket Name	Result	Comments
aws-cloudtrail-logs-798094440826-aa81e64c	Fail	Missing tags: ['Classification'].
config-bucket-798094440826	Fail	Tags not found on this bucket.
bad-bucket-54321	Pass	
itauditguy	Pass	
taylor-swift-secret-vault	Pass	

[BACK TO TEST SUMMARY](#)

Test ID	aws-ec2-001
Test Description	EC2 instances must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained the list of EC2 instances by calling describe_instances() boto3 command. 2. For each in-scope AWS region, saved the list of EC2 instances: ec2/[region_name]/instances.json 3. For each EC2 instance, reviewed the 'Tags' to determine if the following tag keys exist and have non-empty values: ['Owner', 'Description', 'Classification']
Conclusion	Fail
Comments	Exceptions Noted. 1 of 1 EC2 instance(s) are missing required tags or have empty values.

Region	Instance ID	Result	Comments
us-east-1	i-00ad64923065104d8	Fail	Missing tags: ['Owner', 'Description', 'Classification'].

BACK TO TEST SUMMARY

Test ID	aws-ec2-001
Test Description	EC2 security groups have required tags applied and tag values are not empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of EC2 security groups by calling describe_security_groups() boto3 command. 2. For each in-scope region, saved the list of security groups: ec2/[region]/security_groups.json 3. Inspected each security group's 'Tags' attribute to determine if the following tag keys exist and have non-empty values: ['Owner', 'Description', 'ReviewedBy', 'LastReviewedDate']
Conclusion	Fail
Comments	Exceptions Noted. 3 of 3 security group(s) are missing required tags or have empty values.

Region	Security Group ID	Result	Comments
us-east-1	sg-0d3a9f0b108d87954	Fail	Missing tags: ['ReviewedBy', 'LastReviewedDate'].
us-east-1	sg-0c8fdfe934b59f4fa	Fail	Missing tags: ['Owner', 'Description', 'ReviewedBy', 'LastReviewedDate'].
us-east-1	sg-0f1c8a3d07e49bb15	Fail	Missing tags: ['Owner', 'Description', 'ReviewedBy', 'LastReviewedDate'].

BACK TO TEST SUMMARY

Test ID	aws-ebs-003
Test Description	EBS volumes must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained the list of EBS volumes by calling describe_volumes() boto3 command. 2. Saved the list of volumes in the audit evidence folder (ec2/[region_name]/volumes.json). 3. For each volume, obtained its tags from the 'Tags' attribute. 4. Inspected each EBS volume to determine if the following tag keys exist and have non-empty values: ['Owner', 'Description', 'Classification']
Conclusion	Fail
Comments	Exceptions Noted. 1 of 1 EBS volume(s) are missing required tags or have empty values.

Region	Volume ID	Result	Comments
us-east-1	vol-0fad5b6c1c9198060	Fail	Missing tags: ['Owner', 'Description', 'Classification'].

BACK TO TEST SUMMARY

Test ID	aws-lambda-001
Test Description	Lambda functions must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained the list of Lambda functions by calling list_functions() boto3 command. 2. Saved the list of functions in the audit evidence folder (lambda/[region_name]/functions.json). 3. For each function, obtained its tags using list_tags() boto3 command. 4. Saved the tags for each function in the audit evidence folder (lambda/[region_name]/functions/[function_name]/tags.json). 5. Inspected each Lambda function to determine if the following tag keys exist and have non-empty values: ['Owner', 'Description', 'Classification']
Conclusion	Fail
Comments	Exceptions Noted. 3 of 3 Lambda function(s) are missing required tags or have empty values.

Region	Function Name	Result	Comments
us-east-1	upload_audit_evidence	Fail	Missing tags: ['Owner', 'Description', 'Classification'].
us-east-1	build_student_accounts	Fail	Missing tags: ['Owner', 'Description', 'Classification'].
us-east-1	aws_training_backend	Fail	Missing tags: ['Owner', 'Description', 'Classification'].

BACK TO TEST SUMMARY

Test ID	aws-s3-003
Test Description	S3 buckets are configured to deny unencrypted data in-transit.
Risk Rating	Informational
Test Procedures	1. Obtained a list of S3 buckets by calling the list_buckets() boto3 command. 2. Saved the list of buckets: s3/buckets.json. 3. For each bucket, obtained the bucket policy by calling the get_bucket_policy() boto3 command. 4. For each bucket, saved the bucket policy: s3/buckets/[bucket_name]/bucket_policy.json. 5. For each bucket, inspected the bucket policy to determine if a statement exists that denies requests when aws:SecureTransport is false.
Conclusion	Fail
Comments	Exceptions Noted. 4 of 5 S3 bucket(s) do not enforce secure transport (HTTPS).

Bucket Name	Result	Comments
bad-bucket-54321	Fail	No bucket policy statement enforcing SecureTransport.
config-bucket-798094440826	Fail	No bucket policy statement enforcing SecureTransport.
itauditguy	Fail	No bucket policy statement enforcing SecureTransport.
taylor-swift-secret-vault	Fail	No bucket policy statement enforcing SecureTransport.
aws-cloudtrail-logs-798094440826-aa81e64c	Pass	

BACK TO TEST SUMMARY

Test ID	aws-iam-001
Test Description	Root account does not have any active access keys.
Risk Rating	High
Test Procedures	1. Obtained the AWS account summary by calling the get_account_summary() boto3 command. 2. Saved the account summary: iam/account_summary.json. 3. Inspected the account summary to determine if 'AccountAccessKeysPresent' is set to 0.
Conclusion	Pass

BACK TO TEST SUMMARY

Test ID	aws-iam-003
Test Description	Root account has MFA enabled.
Risk Rating	High
Test Procedures	1. Obtained the AWS account summary by calling the get_account_summary() boto3 command. 2. Saved the account summary: iam/account_summary.json 3. Inspected the account summary to determine if 'AccountMFAEnabled' is set to 1.
Conclusion	Pass

BACK TO TEST SUMMARY

Test ID	aws-rds-002
Test Description	RDS instances are configured to block public access.
Risk Rating	High
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling the describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json. 3. For each RDS instance, inspected the 'PubliclyAccessible' setting to determine if it was set to 'false'.
Conclusion	Pass

Region	DB Instance	Result	Comments
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BACK TO TEST SUMMARY

Test ID	aws-cloudtrail-001
Test Description	At least one multi-region CloudTrail trail has logging enabled.
Risk Rating	High
Test Procedures	1. Obtained a list of CloudTrail trails by calling the describe_trails() boto3 command. 2. Saved the list of CloudTrail trails: cloudtrail/trails.json. 3. For each CloudTrail trail, inspected the trail configuration to determine whether 'IsMultiRegionTrail' is set to 'true'. 4. For each multi-region trail, obtained the trail status by calling the get_trail_status() boto3 command. 5. For each multi-region trail, saved the trail status: cloudtrail/trails/[trail_name]/trail_status.json. 6. Inspected the trail configuration and status to determine if at least one trail complies with the test attribute(s) defined below.
Test Attributes	• At least one trail must have IsMultiRegionTrail = true and IsLogging = true.
Conclusion	Pass

BACK TO TEST SUMMARY

Test ID	aws-guardduty-001
Test Description	GuardDuty is enabled for all in-scope regions.
Risk Rating	High
Test Procedures	1. For each in-scope region, obtained a list of GuardDuty detectors by calling the list_detectors() boto3 command. 2. For each in-scope region, saved the list of detector IDs: guardduty/[region]/detectors.json. 3. For each detector ID, obtained detector configuration by calling the get_detector() boto3 command. 4. For each detector ID, saved the detector configuration: guardduty/[region]/[detector_id]/config.json. 5. For each detector ID, inspected the detector configuration to determine whether 'Status' is set to 'ENABLED'.
Conclusion	Pass

Region	Result	Comments
us-east-1	Pass	

BACK TO TEST SUMMARY

Test ID	aws-iam-006
Test Description	IAM passwords comply with the organization's password policy.
Risk Rating	Medium
Test Procedures	1. Obtained the IAM password policy by calling the <code>get_account_password_policy()</code> boto3 command. 2. Saved the password policy: <code>iam/password_policy.json</code>. 3. Inspected the password policy to determine if it complies with the configured requirements.
Test Attributes	• Minimum password length ≥ 14. • Password history ≥ 24. • Passwords require symbols. • Passwords require numbers. • Passwords require uppercase letters. • Passwords require lowercase letters.
Conclusion	Pass

BACK TO TEST SUMMARY

Test ID	aws-s3-001
Test Description	S3 buckets are encrypted at rest.
Risk Rating	Medium
Test Procedures	1. Obtained a list of S3 buckets by calling the <code>list_buckets()</code> boto3 command. 2. Saved the list of buckets: <code>s3/buckets.json</code>. 3. For each S3 bucket, obtained the encryption settings by calling the <code>get_bucket_encryption()</code> boto3 command. 4. For each S3 bucket, saved the encryption settings: <code>s3/buckets/[bucket_name]/encryption.json</code>. 5. For each S3 bucket, inspected the encryption settings to determine if they comply with the test attribute(s) below.
Test Attributes	• <code>ServerSideEncryptionConfiguration</code> is present.
Conclusion	Pass

Bucket Name	Result	Comments
aws-cloudtrail-logs-798094440826-aa81e64c	Pass	
bad-bucket-54321	Pass	

config-bucket-798094440826	Pass	
itauditguy	Pass	
taylor-swift-secret-vault	Pass	

BACK TO TEST SUMMARY

Test ID	aws-rds-001
Test Description	RDS instances are encrypted at rest.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling the describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json. 3. For each RDS instance, inspected the `StorageEncrypted` setting to determine if it was set to `true`.
Conclusion	Pass

Region	DB Instance	Result	Comments
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BACK TO TEST SUMMARY

Test ID	aws-rds-005
Test Description	RDS instances have deletion protection enabled at the cluster or instance level.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained a list of RDS instances and RDS clusters using describe_db_instances() and describe_db_clusters() boto3 commands. 2. Saved the list of RDS instances: rds/[region_name]/db_instances.json and DB clusters: rds/[region_name]/db_clusters.json. 3. Inspected each RDS instance to determine if 'DeletionProtection' was set to 'true' at the instance or cluster level.
Conclusion	Pass

Region	DB Instance	Result	Comments
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BACK TO TEST SUMMARY

Test ID	aws-rds-004
Test Description	RDS instances have automatic minor version upgrades enabled.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of DB instances by calling the describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json. 3. For each RDS instance, inspected the 'AutoMinorVersionUpgrade' setting to determine if it was set to 'true'.
Conclusion	Pass

Region	DB Instance	Result	Comments
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BACK TO TEST SUMMARY

Test ID	aws-rds-003
Test Description	RDS backups are retained for at least 14 days.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling the <code>describe_db_instances()</code> boto3 command. 2. For each in-scope region, saved the list of RDS instances: <code>rds/[region_name]/db_instances.json</code>. 3. For each RDS instance, inspected the <code>BackupRetentionPeriod`</code> to determine if it is greater than or equal to 14 days.
Conclusion	Pass

Region	DB Instance	Result	Comments
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BACK TO TEST SUMMARY

Test ID	aws-rds-006
Test Description	RDS instances must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json. 3. For each RDS instance, reviewed the `TagList` to determine if the following tag keys exist and have non-empty values: ['Owner', 'Description', 'Classification']
Conclusion	Pass

Region	DB Instance	Result	Comments
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